Auditing the Costs of Multi-Group Fairness Repair

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Equal opportunity (EO) requires a classifier to have the same true-positive rate (TPR) across demographic groups. When fairness concerns multiple demographic attributes, repairing EO for each attribute separately does not ensure EO across their intersections. We study three published postprocessing methods for intersectional EO on two datasets, Credit Default and COMPAS, focusing on accuracy loss and withdrawals of previously favorable predictions. We then introduce Retain-EG, which modifies one of these methods, exponentiated gradient (EG), to reduce withdrawals while limiting changes in accuracy and the TPR gap. The baseline methods achieve near-zero intersectional TPR gaps on the repair data, but generalize poorly on COMPAS, where test gaps remain substantial. Some methods achieve fitted parity at the cost of large accuracy losses and high withdrawal rates. Retain-EG can reduce test withdrawals with small accuracy losses in some fits, but these gains do not eliminate the substantial remaining disparities or uneven group-level costs. Code is available at https://github.com/Xin-Lyra-Zheng/DS-709-Final-Project.git.

Additional Key Words and Phrases: equal opportunity, intersectional fairness, postprocessing, decision withdrawals

1 Introduction

Equal opportunity (EO) requires a classifier to have the same true-positive rate across demographic groups, so that individuals with a favorable true label have an equal chance of receiving a favorable prediction regardless of group membership. For a trained classifier, EO can be pursued through postprocessing, which adjusts decision thresholds or prediction probabilities across groups. When several demographic attributes matter, enforcing EO for each attribute separately may leave disparities among their intersections. Intersectional EO addresses this by applying the fairness constraints to groups defined jointly by these attributes, such as race and sex. However, satisfying these constraints does not by itself ensure that a repair avoids unnecessary accuracy loss or withdrawals of previously favorable predictions.

Consider eight positive-label applicants in two groups of four in Figure 1. An original classifier approves three in one group and one in the other. Two repairs can approve two per group, giving identical TPRs and accuracy, while retaining different original approvals. One withdraws a single approval; the other withdraws two and compensates by approving previously rejected applicants. Their withdrawal rates are therefore 25% and 50%. Figure 1 illustrates why final fairness and accuracy do not identify decision reversals.

We ask how existing methods for intersectional EO affect predictive accuracy and withdraw previously favorable predictions, and whether these costs can be reduced while meeting the fairness target. We evaluate accuracy and withdrawals as two distinct outcomes: withdrawing an incorrect favorable prediction improves accuracy, whereas withdrawing a correct one reduces it. The audit treats these outcomes as equally relevant costs of fairness repair. Our extension then tests one concrete intervention: reducing withdrawals while placing explicit limits on changes in fairness and accuracy.

We audit three methods on Credit Default and COMPAS using the same logistic-regression (LR) and histogram-gradient-boosting (HGB) scores: Hardt–Price–Srebro postprocessing [4], Morina et al.'s RTDP [6], and Agarwal et al.'s exponentiated-gradient (EG) reduction [1]. Our audit examines TPR gaps for individual sensitive attributes and their intersections, accuracy loss, and withdrawals of original positive predictions. We assess accuracy and withdrawals both overall and within groups, and compare all three dimensions on repair and test data. The results reveal trade-offs between fairness, accuracy, and withdrawals, with costs that are unevenly distributed across groups. They also show that intersectional parity achieved on the repair data can leave substantial disparities on unseen data.

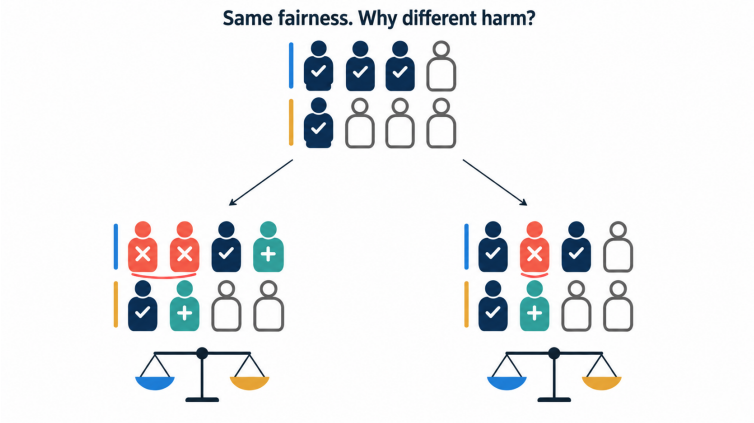


Fig. 1. Two repairs achieve the same EO and accuracy but withdraw different numbers of original approvals.

Building on this audit, we introduce Retain-EG to reduce the costs of intersectional fairness repair. Retain-EG adjusts the weights of EG's existing classifiers to reduce withdrawals while limiting changes in fairness and accuracy on both repair and validation data. It reduces test withdrawals across all twenty model–split fits, with nearly unchanged average accuracy. However, improvements in test fairness vary, and some groups still experience accuracy losses. Our study combines an audit of the accuracy and withdrawal costs of existing EO methods with a targeted modification that reduces withdrawals while preserving average predictive performance.

2 Related Work

Fairness across individual demographic attributes does not necessarily protect their intersections. Kearns et al. [5] demonstrate this limitation and develop methods for protecting richer families of subgroups. This motivates evaluating EO across groups defined jointly by multiple sensitive attributes, rather than checking each attribute separately. Several methods can enforce these constraints. Hardt et al. [4] introduce postprocessing for equal opportunity and equalized odds by adjusting a trained predictor. Morina et al. [6] address intersectional fairness through group-specific thresholds and randomized predictions. Agarwal et al. [1] reduce fairness-constrained classification to cost-sensitive learning and produce a mixture of classifiers. We use these methods as baselines, applying their EO constraints to explicitly defined intersections. However, satisfying empirical constraints does not ensure fairness on new data. Cotter et al. [2] study constraint generalization using separate samples for model and constraint learning. Their work motivates our use of validation data when modifying a fitted repair, alongside a comparison of repair-set and test-set outcomes.

Field Landscape. Within AI ethics, algorithmic fairness is part of a broader concern with how prediction models are evaluated and deployed. EO specifies one criterion for comparing groups, but it's very importnat to ask who benefits from a system and who bears its costs. Our audit connects these perspectives by examining how fairness repairs change accuracy and previously favorable predictions across groups. The project also relates to deployment monitoring: fairness measured during policy development may not persist on new data. These connections place intersectional EO within the wider task of evaluating both the distribution of a system's costs and the reliability of its behavior.

3 Methods

3.1 Baseline methods

Let X denote the features, $Y \in \{0,1\}$ the true label, and $A = (A_1, \ldots, A_K)$ the sensitive attributes. A randomized policy approves a record with probability $q(X, A)$. EO requires $\Pr(\widehat{Y} = 1 \mid Y = 1, G = g)$ to be equal across groups g; equivalently, $r_g(q) = \mathbb{E}[q(X, A) \mid Y = 1, G = g]$ is constant. For marginal EO, G is one attribute A_k; for intersectional EO, G is the full tuple A. Equality after averaging over each attribute separately need not imply equality within every tuple. Let $\mathcal{G}$ be the fitted intersection categories. We fit the three baselines on the repair sample R, using a frozen score $s(X)$ and intersectional group membership.

Hardt–Price–Srebro postprocessing. Hardt et al. [4] select group-specific thresholds and randomize between threshold rules to meet EO while maximizing accuracy. Randomization fills gaps between operating points that deterministic thresholds cannot attain. Writing $h_{g,j}(x) = \mathbf{1}\{s(x) \geq t_{g,j}\}$, the policy and empirical optimization are

$$q_\alpha(x, g) = \sum_j \alpha_{g,j} h_{g,j}(x), \qquad \max_{\alpha_g \in \Delta} \widehat{\mathrm{Acc}}_R(q_\alpha) \quad \text{s.t. } \widehat{r}_{g,R}(q_\alpha) = \widehat{r}_{h,R}(q_\alpha) \quad \forall g, h. \tag{1}$$

Here Δ is the probability simplex, $\widehat{r}_{g,R}$ is the mean approval probability among group g's positive-label repair records, and $\widehat{\mathrm{Acc}}_R$ is expected repair accuracy. Both rates are linear in the mixture weights. Geometrically, the method chooses compatible points on the groups' convexified threshold ROC curves. The fitted optimum is over these score-derived policies; it does not establish parity on unseen data. We use Fairlearn's threshold optimizer with TPR parity and accuracy maximization; fitted cells must contain both labels, and other cells retain the original threshold.

Morina RTDP.. Morina et al. [6] combine a group-specific threshold with separate approval probabilities above and below it. We use their sequential optimization: first choose each threshold to maximize that group's repair accuracy, then hold the thresholds fixed while fitting the probabilities. Formally,

$$\begin{aligned} \tau_g &\in \arg\max_t \widehat{\mathrm{Acc}}_{g,R}(\mathbf{1}\{s(X) \geq t\}), \\ q_p(x, g) &= p_{g,1}\mathbf{1}\{s(x) \geq \tau_g\} + p_{g,0}\mathbf{1}\{s(x) < \tau_g\}, \\ \max_{p \in [0,1]^{2|\mathcal{G}|}} & \quad \widehat{\mathrm{Acc}}_R(q_p) \quad \text{s.t. } \widehat{r}_{g,R}(q_p) = \widehat{r}_{h,R}(q_p) \quad \forall g, h. \end{aligned} \tag{2}$$

The second stage is a linear program. Records on the same side of a group's threshold share an approval probability, so the policy can grant new approvals as well as withdraw existing ones. A constant approval probability makes the parity constraints feasible, but fixing thresholds before enforcing fairness can exclude a better joint choice of thresholds and probabilities. Thus optimality of this second-stage program does not imply optimality over all RTDP policies.

Agarwal EG.. Agarwal et al. [1] turn fair classification into repeated cost-sensitive learning, increasing the costs of decisions that violate the constraints. The output combines the resulting classifiers rather than choosing a single rule. Let $h_1, \ldots, h_M$ be candidate rules, $w \in \Delta_M$ their weights, and $\widehat{\gamma}(q) \leq d$ the linear empirical fairness constraints. The mixture and Lagrangian are

$$q_w(x, g) = \sum_{t=1}^{M} w_t h_t(x, g), \qquad L(w, \lambda) = 1 - \widehat{\mathrm{Acc}}_R(q_w) + \lambda^\top(\widehat{\gamma}(q_w) - d), \quad \lambda \geq 0. \tag{3}$$

EG updates the constraint multipliers by exponentiated gradient and calls an oracle to find a classifier minimizing the current cost. Its theoretical approximation applies to constrained empirical error minimization under the stated oracle

and convergence assumptions, not to exact population EO. Our oracle searches group-specific thresholds over frozen scores, making this application postprocessing. The returned rules and original weights w^0 are saved for Retain-EG.

The methods differ in what they optimize: Hardt searches randomized threshold operating points, our RTDP configuration fixes a threshold before optimizing two probabilities, and EG discovers rules through cost-sensitive optimization and combines them. Hardt and RTDP solve linear programs over their chosen policies; EG uses an iterative reduction and final mixture refinement. Their policy classes, constraint implementations, and coverage rules differ, so no general accuracy ordering follows from the shared EO goal.

None of these objectives directly penalizes reversals of the original classifier's favorable predictions. Their different policy classes and coverage rules can also affect the observed costs. We therefore compare attained fairness, accuracy, and withdrawals without treating the methods as optimizers over an identical feasible set. Appendix A gives the implementation settings.

3.2 Data and experimental design

We use UCI Credit Default [8] and COMPAS [7]. Credit contains 30,000 records; the favorable label is no default next month, and the sensitive attributes are sex and age below 40 versus at least 40. The base models exclude ID, sex, age, and the target column. COMPAS contains 6,172 records after filtering, with no recorded two-year recidivism as the favorable label. We audit race, sex, and their intersections, using prior-count, juvenile-count, and charge-degree features. These are our fitted models' predictions, not the proprietary COMPAS score; favorable predictions represent hypothetical decisions.

For each dataset, five seeds create disjoint training, calibration, repair, validation, and test partitions in proportions 50/10/15/10/15. We train LR and HGB, apply sigmoid calibration, and freeze the scores. The original classifier's threshold maximizes repair-set accuracy. All methods share the scores, original predictions, and partitions, yielding twenty paired model–split fits. Repair data fit the fairness methods, validation data constrain Retain-EG, and test data evaluate final outcomes. Records recur across splits, so comparisons are descriptive rather than independent replications.

For the reported intersectional comparisons, we include only groups containing at least twenty records with $Y = 1$ in the repair set. TPR is estimated from these positive-label records; this cutoff avoids reporting rates based on very small denominators. It retains all four Credit intersections and six COMPAS intersections: African-American and Caucasian women and men, Hispanic men, and Other men. Groups below the cutoff are excluded from the headline intersectional summaries, not from the dataset. Marginal comparisons cover each sensitive attribute separately. Appendix A gives partition sizes and filtering details..

3.3 Evaluation metrics

We audit EO attainment, accuracy loss, and withdrawals on both repair and test data. For partition D, let D_g contain group g's records, $D_g^+ = \{i \in D_g : Y_i = 1\}$, $b_i \in \{0, 1\}$ be the original prediction, and q_i the repaired approval probability. The family $\mathcal{F}$ contains either one attribute's categories or the reported intersections.

TPR gap. This measures how closely the repair achieves EO and whether fitted parity transfers to test data:

$$\widehat{\mathrm{TPR}}_{g,D}(q) = \frac{\sum_{i \in D_g^+} q_i}{|D_g^+|}, \qquad \widehat{\mathrm{Gap}}_{\mathcal{F},D}(q) = \max_{g \in \mathcal{F}} \widehat{\mathrm{TPR}}_{g,D}(q) - \min_{g \in \mathcal{F}} \widehat{\mathrm{TPR}}_{g,D}(q). \tag{4}$$

We report marginal and intersectional gaps separately because they compare different groups.

Accuracy loss. This measures the predictive cost of the repair relative to the original classifier, overall and within each group:

$$\widehat{\mathrm{Acc}}_{g,D}(q) = \frac{1}{|D_g|} \sum_{i \in D_g} [Y_i q_i + (1 - Y_i)(1 - q_i)],$$

$$\Delta \mathrm{Acc}_{g,D}(q; b) = \widehat{\mathrm{Acc}}_{g,D}(b) - \widehat{\mathrm{Acc}}_{g,D}(q). \tag{5}$$

The overall definition uses D in place of D_g; a negative loss means improvement. We also report worst-group accuracy, $\min_{g \in \mathcal{F}_\cap} \widehat{\mathrm{Acc}}_{g,D}(q)$, where $\mathcal{F}_\cap$ is the intersection family. It describes the lowest final performance, rather than the largest group-specific loss.

Withdrawals. This measures decision reversals that accuracy alone cannot identify:

$$\widehat{W}_{g,D}(q; b) = \frac{\sum_{i \in D_g} b_i (1 - q_i)}{\sum_{i \in D_g} b_i}. \tag{6}$$

We use the same definition overall and report the maximum intersectional group rate. The denominator contains all original approvals, including incorrect ones, whereas the false-negative rate conditions on $Y = 1$. Withdrawing a correct approval lowers accuracy; withdrawing an incorrect one improves it. We therefore audit accuracy and withdrawals as distinct costs under the EO target.

All metrics use approval probabilities, averaging over prediction randomization. Repair-to-test changes therefore reflect sampling and policy generalization, not simulated binary draws. Worst-group accuracy and maximum-group withdrawal are computed within each fit before averaging.

3.4 Retain-EG

EG's final mixture can contain multiple ways to attain similar fairness and accuracy while retaining different original approvals. Retain-EG keeps its saved rules $h_1, \ldots, h_M$ fixed and changes only the weights. It minimizes withdrawals while limiting changes in EO attainment and overall accuracy on repair and validation data.

Let R and V be those two partitions, w^0 the original EG weights, and $q_w = \sum_t w_t h_t$. For each audited family and partition, set $\tau_{\mathcal{F},D} = \max\{\epsilon, \widehat{\mathrm{Gap}}_{\mathcal{F},D}(q_{w^0})\}$. We solve

$$\begin{aligned} \min_{w \in \Delta_M} \quad & \widehat{W}_R(q_w; b) \\ \text{subject to} \quad & \widehat{\mathrm{Gap}}_{\mathcal{F},D}(q_w) \le \tau_{\mathcal{F},D} \quad \forall \mathcal{F},\ D \in \{R, V\}, \\ & \widehat{\mathrm{Acc}}_D(q_w) \ge \widehat{\mathrm{Acc}}_D(q_{w^0}) - \delta \quad D \in \{R, V\}, \\ & \|w - w^0\|_1 \le \rho. \end{aligned} \tag{7}$$

We use $\epsilon = .02$, $\delta = .005$, and $\rho = .25$. A gap below two percentage points may rise to that floor; a larger gap cannot worsen on the constrained partition. Accuracy may fall by at most half a point, and the weight bound limits the intervention. The objective and accuracy are linear in w, and the gap bound is equivalent to linear pairwise TPR inequalities, so this is a linear program. The original weights remain feasible, guaranteeing no increase in optimized repair-set withdrawals up to numerical tolerance. These empirical constraints do not guarantee test fairness or accuracy. Appendix B gives the coefficient form; model and subgroup results appear in Appendix C.

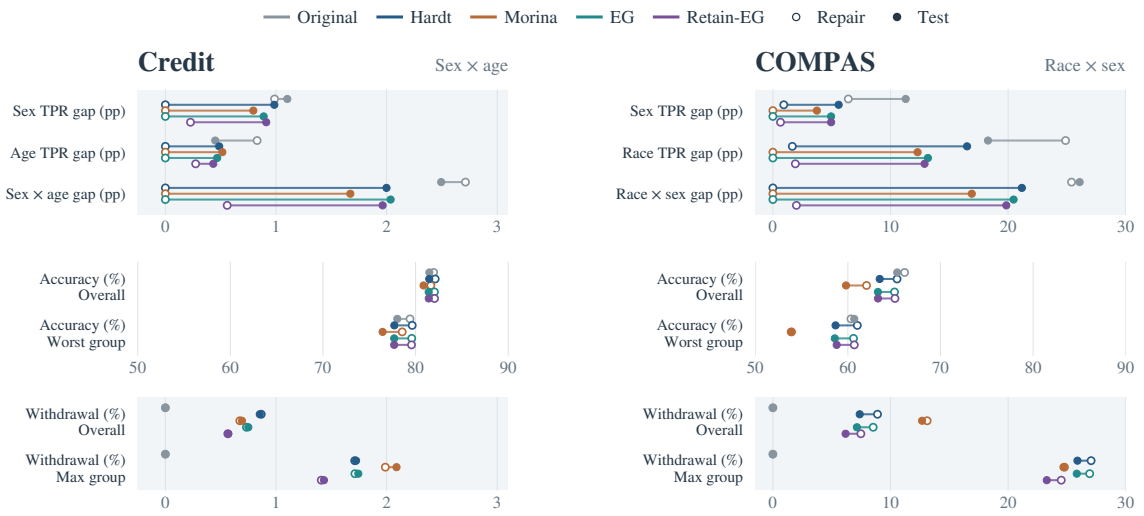


Fig. 2. Fairness, accuracy, and withdrawals on Credit Default and COMPAS. Open and filled points show repair-set and test-set means over two models and five splits. Lines connect the two partitions. Accuracy is absolute; withdrawal is measured relative to original favorable predictions. Worst-group accuracy and maximum-group withdrawal are computed within each fit before averaging. Each metric has its own axis.

4 Results

4.1 Baseline audit

The audit plot in Figure 2 compares marginal and intersectional TPR gaps, accuracy, and withdrawals on repair and test data. Credit starts with relatively small disparities: the baselines bring modest test-gap reductions with small accuracy changes and few withdrawals. COMPAS starts with much larger disparities, making the costs and generalization limits more visible.

On COMPAS, RTDP achieves the smallest mean test intersectional gap but also the largest overall accuracy loss and withdrawal rate. Hardt and EG preserve more accuracy and original approvals while leaving larger test gaps. Marginal gaps are smaller than intersectional gaps, so auditing individual attributes alone would understate the remaining disparities. The plot also shows that lower overall accuracy is accompanied by lower worst-group accuracy. These minima identify the weakest final performance, but do not measure each group's loss from its own original accuracy. Withdrawals are concentrated within particular intersections: maximum-group rates substantially exceed overall rates. The method with the largest overall withdrawal rate does not have the largest maximum-group rate. Aggregate costs therefore cannot identify the groups most affected by a repair.

All three methods achieve approximately zero intersectional gaps on the repair data, yet sizable gaps remain on the COMPAS test data. Marginal gaps also persist, while the accuracy and withdrawal panels show that their costs carry into test evaluation. Fitted parity thus gives an incomplete account of a repair's performance. Sparse positive-label records may contribute to unstable group TPR estimates, but this audit does not isolate their effect from changes in scores or policy selection.

4.2 Retain-EG

Retain-EG reduces test withdrawals in all twenty paired fits, with nearly unchanged average accuracy and smaller mean intersectional gaps (Table 1). These averages do not imply consistent improvement: all three outcomes improve or tie in only half the fits on each dataset. Group-level costs remain uneven. Some groups lose accuracy relative to EG, and a Credit group experiences more withdrawals despite the overall reduction. Retaining more original approvals preserves both correct and incorrect predictions, so fewer withdrawals need not improve accuracy. Appendix C reports the group, model, and label breakdowns.

Table 1. Test outcomes of Retain-EG relative to EG, averaged over two models and five splits per dataset. Gaps and accuracy changes are in percentage points (pp); withdrawals are percentages of original favorable predictions. The final column counts fits with no larger intersectional gap, no lower accuracy, and no greater withdrawal.

Dataset	EG withdrawal	Retain-EG withdrawal	Δ gap (pp)	Δ accuracy (pp)	All three improve or tie
Credit Default	0.750%	0.569%	−0.07	$\approx +0.01$	5/10
COMPAS	7.144%	6.181%	−0.62	$\approx +0.01$	5/10

5 Discussion

The audit addresses the gap between attaining EO and understanding its costs. Lower TPR gaps can accompany larger accuracy losses and more withdrawals, while identical fairness and accuracy can involve different decision reversals. These outcomes must therefore be evaluated together, with accuracy and withdrawals treated as distinct costs under the fairness target.

Retain-EG shows that changing an existing mixture can reduce withdrawals while keeping average accuracy nearly unchanged. Because it permits fitted fairness slack, this result does not establish improvement at an identical fairness target. Whether the exchange is acceptable depends on the application's EO requirement. Group-level results also limit the aggregate conclusion: overall accuracy and withdrawals can improve while a particular group's accuracy falls or withdrawals increase. The current objective and accuracy constraint do not protect every group separately.

Generalization remains unresolved. Validation constraints limit changes on an additional sample but do not guarantee parity on new records. Small intersections make TPR estimation difficult, and excluding groups with insufficient support limits coverage. The same records recur across splits, and test results were inspected during development, so the paired comparisons are descriptive. Finally, these benchmark labels and hypothetical decisions support an audit of prediction changes, rather than conclusions about their causal welfare effects.

6 Conclusion

Intersectional EO should be evaluated together with its accuracy and withdrawal costs, both overall and within groups. Our audit finds substantial differences among existing methods and shows that fitted parity can leave large test disparities. Retain-EG reduces withdrawals across all twenty inspected fits with nearly unchanged average accuracy, but its benefits depend on the allowed fairness slack and do not extend uniformly to every subgroup. The results support making both costs explicit when comparing fairness repairs and checking whether the observed balance persists on new data.

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A Implementation Details

A.1 Datasets and labels

We use UCI Credit Default [8] and COMPAS [7], which differ in their initial group disparities. Credit contains 30,000 records. The favorable label is no default next month, and the sensitive attributes are sex and age below 40 versus at least 40, yielding four intersections. The base models exclude ID, sex, age, and the target column; payment history and financial features remain. COMPAS contains 6,172 records after the pipeline's screening-window, recidivism-availability, charge-degree, and score-availability filters. The favorable label is no recorded two-year recidivism. We audit race, sex, and their intersections, using prior-count, juvenile-count, and charge-degree features. These are predictions from our fitted models, rather than the proprietary COMPAS score, and favorable predictions are hypothetical decisions.

A.2 Partitions and group support

Seeds 709–713 produce training, calibration, repair, validation, and test partitions of 15,000, 3,000, 4,500, 3,000, and 4,500 records for Credit; the corresponding COMPAS counts are 3,086, 617, 926, 617, and 926. Headline intersections must have at least twenty repair-set positives. This retains four Credit and six COMPAS intersections in every fit. Full counts and unfiltered group metrics should accompany the experiment artifacts.

A.3 Baseline configurations

Hardt randomized thresholds. We use the threshold optimizer in Fairlearn 0.12 [3], with TPR parity, accuracy maximization, a 1,000-point grid, and `flip=False`. Race–sex or sex–age intersections are encoded as categories. A fitted cell must contain both labels; other cells retain the original threshold. This coverage rule is reported because it can leave nonzero marginal gaps even when the supported intersectional gap is zero.

Morina RTDP.. For each intersection g, we first choose an accuracy-maximizing threshold τ_g from observed repair scores. We then hold it fixed and solve a linear program for $p_{g,0}, p_{g,1} \in [0, 1]$, giving

$$q(x) = p_{g,1}\mathbf{1}\{s(x) \geq \tau_g\} + p_{g,0}\mathbf{1}\{s(x) < \tau_g\}. \tag{8}$$

The objective is expected accuracy, with equal fitted TPRs in cells containing positives. Thus the method can grant as well as withdraw approvals. Same-group records on the same side of the threshold receive the same probability. Missing cells use an explicitly documented pooled fallback. This is the unregularized two-stage configuration, not the project's earlier pooled-penalty extension.

Agarwal EG.. We use Fairlearn's `ExponentiatedGradient` with TPR-parity moments, at most fifty iterations, `eps=.005`, `nu=1e-5`, and its linear-program refinement. The cost-sensitive oracle searches group-specific thresholds on a 0.01 score grid. The result is a mixture of deterministic rules over frozen scores, with $q_{w^0}(x) = \sum_t w_t^0 h_t(x)$. The dictionaries contain 14–78 returned rules in this audit. EG is a reduction framework, not intrinsically a postprocessor; our oracle makes this application postprocessing. Different baseline policy classes and coverage rules prevent a class-matched superiority interpretation.

B Retain-EG Linear Program

B.1 Linear program

Retain-EG keeps EG's rules fixed and changes only their mixture weights. For repair sample R, define $H_{it} = h_t(x_i)$ and $q_w = Hw$, with w in the probability simplex Δ_M. The overall withdrawal objective is linear in the weights:

$$\widehat{W}_R(w) = c^\top w, \qquad c_t = \frac{\sum_{i \in R} b_i (1 - H_{it})}{\sum_{i \in R} b_i}. \tag{9}$$

The audit gives accuracy and withdrawals equal attention; this extension specifically minimizes withdrawals while constraining accuracy loss. It does not optimize either outcome separately for every group.

Let $a_{g,D,t}$ be rule t's mean prediction among positive-label records in group g on partition D, and let $u_{D,t}$ be its overall accuracy. For each audited family, set $\tau_{\mathcal{F},D} = \max\{\epsilon, \widehat{\mathrm{Gap}}_{\mathcal{F},D}(w^0)\}$. We solve

$$\begin{aligned} \min_{w \in \Delta_M} \quad & c^\top w \\ \text{subject to} \quad & (a_{g,D} - a_{h,D})^\top w \le \tau_{\mathcal{F},D}, \\ & \qquad g, h \in \mathcal{F}, \quad D \in \{R, V\}, \\ & u_D^\top w \ge u_D^\top w^0 - \delta, \quad D \in \{R, V\}, \\ & \|w - w^0\|_1 \le \rho. \end{aligned} \tag{10}$$

Here V is the validation partition. We use $\epsilon = .02$, $\delta = .005$, and $\rho = .25$. Thus a baseline gap below two percentage points may increase to two points, overall accuracy may fall by at most half a point on either fitting partition, and the weight change is bounded. If EG's validation gap already exceeds two points, the cap prevents further worsening of that measured gap. These are empirical constraints, not guarantees for new data.

The original weights are feasible, so the optimized repair-set withdrawal cannot increase. Validation constraints restrict changes on data not used to generate the rules, while the weight bound limits the size of the intervention.

C Model and Subgroup Results

The gain is present with both score models. COMPAS test withdrawal falls from 8.158% to 7.202% with LR and from 6.130% to 5.159% with HGB; corresponding Credit reductions are 0.963% to 0.737% and 0.537% to 0.401%. Accuracy increases with LR but decreases slightly with HGB on both datasets. Aggregation therefore hides a model-dependent trade-off.

Label-wise counts reveal another trade-off. Per 926-record COMPAS test partition, Retain-EG produces a mean net increase of 3.46 retained originally correct approvals and 1.99 retained originally incorrect approvals relative to EG. Thus only 63.5% of the additional retained approvals are correct under the benchmark label; the corresponding Credit share is 52.8%. Retain-EG also creates new approvals: on COMPAS it adds 2.42 true positives and 3.80 false positives relative to EG. These opposing changes explain how a visible withdrawal reduction can coexist with nearly unchanged overall accuracy.

Benefits also differ by subgroup. COMPAS's mean withdrawal reduction is 1.78 points for Caucasian women and 2.17 for Other men, but essentially zero for African-American men. The latter group's accuracy falls by 0.32 points: its additional false-positive approvals outweigh additional true positives. Across the supported subgroup–fit comparisons, accuracy decreases in 15/60 COMPAS cases and 11/40 Credit cases. In one Credit fit, women aged 40+ also have

a 0.40-point withdrawal increase despite the overall reduction. An aggregate retention objective therefore leaves a concrete subgroup-protection gap; the mean improvement is not a no-harm constraint.

D Interpreting Withdrawals

Conditioning on an original positive prediction selects a different population from conditioning on a favorable label. Let $\pi_g = \Pr(Y = 1 \mid b = 1, G = g)$, and let $W_{g,y}$ denote the withdrawal rate among original positives with label y. Then

$$W_g = \pi_g W_{g,1} + (1 - \pi_g) W_{g,0}. \tag{11}$$

Withdrawing an original true positive creates an error, whereas withdrawing an original false positive corrects an error. Consequently, two methods with the same W_g can have different accuracy, and a method with smaller W_g need not be more accurate. The optimization handles this by retaining a separate accuracy constraint instead of treating every reversal as a classification loss. In an application, the classification label may itself be an imperfect proxy for the outcome the decision-maker values, so neither component can automatically be interpreted as causal welfare.